Phase 1

```
aps@apekshaashok:~/AM048_PES_VCS$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
aps@apekshaashok:~/AM048_PES_VCS$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
```

```
aps@apekshaashok:~/AM048_PES_VCS$ find .pes/objects -type f
.pes/objects/f5/902ee73b6aa25436f687067fc4cc0cb481cf416740d3c3c3ed6d9c4de7b6c2
.pes/objects/ad/7fbf2f09e8a48417a997e907713442b93448be515c8fb5741b85877c2e512f
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
aps@apekshaashok:~/AM048_PES_VCS$ xxd .pes/objects/f5/902ee73b6aa25436f687067fc4cc0cb481cf416740d3c3c3ed6d9c4de7b6c2 | head -20
00000000: 7472 6565 2039 3800 3130 3036 3434 2066  tree 98.100644 f
00000010: 696c 6531 2e74 7874 002c f8d8 3d9e e295  ile1.txt.,..=...
00000020: 43b3 4a87 7274 21fd ecb7 e3f3 a183 d337  C.J.rt!........7
00000030: 6390 25de 576d b9eb b431 3030 3634 3420  c.%.Wm...100644
00000040: 6669 6c65 322e 7478 7400 e00c 50e1 6a2d  file2.txt...P.j-
00000050: f38f 8d6b f809 e181 ad02 48da 6e67 19f3  ...k......H.ng..
00000060: 5f9f 7e65 d6f6 0619 9f7f                 _.~e......
```

Phase 2

```
aps@apekshaashok:~/AM048_PES_VCS$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

```
aps@apekshaashok:~/AM048_PES_VCS$ find .pes/objects -type f
.pes/objects/f5/902ee73b6aa25436f687067fc4cc0cb481cf416740d3c3c3ed6d9c4de7b6c2
.pes/objects/ad/7fbf2f09e8a48417a997e907713442b93448be515c8fb5741b85877c2e512f
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
aps@apekshaashok:~/AM048_PES_VCS$ xxd .pes/objects/f5/902ee73b6aa25436f687067fc4cc0cb481cf416740d3c3c3ed6d9c4de7b6c2 | head -20
00000000: 7472 6565 2039 3800 3130 3036 3434 2066  tree 98.100644 f
00000010: 696c 6531 2e74 7874 002c f8d8 3d9e e295  ile1.txt.,..=...
00000020: 43b3 4a87 7274 21fd ecb7 e3f3 a183 d337  C.J.rt!........7
00000030: 6390 25de 576d b9eb b431 3030 3634 3420  c.%.Wm...100644
00000040: 6669 6c65 322e 7478 7400 e00c 50e1 6a2d  file2.txt...P.j-
00000050: f38f 8d6b f809 e181 ad02 48da 6e67 19f3  ...k......H.ng..
00000060: 5f9f 7e65 d6f6 0619 9f7f                 _.~e......
```

Phase 3

```
aps@apekshaashok:~/AM048_PES_VCS$ ./pes init

echo "hello" > file1.txt
echo "world" > file2.txt

./pes add file1.txt file2.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  test_objects.c
  untracked:  pes.h
  untracked:  test_objects
  untracked:  README.md
  untracked:  commit.h
  untracked:  object.c
  untracked:  test_tree.c
  untracked:  index.c
  untracked:  .gitignore
  untracked:  index.h
  untracked:  test_tree
  untracked:  Makefile
  untracked:  commit.c
  untracked:  pes.c
  untracked:  tree.c
  untracked:  test_sequence.sh
  untracked:  tree.h

aps@apekshaashok:~/AM048_PES_VCS$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776747910 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776747910 6 file2.txt
```


```
aps@apekshaashok:~/AM048_PES_VCS$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776747910 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776747910 6 file2.txt
```

Phase 4

```
aps@apekshaashok:~/AM048_PES_VCS$ ./pes log
commit cabfedc966010a7ddc6d57502b7608c9c2768f51fc6d2e645c2d885d5bb2f8cf
Author: Apeksha Ashok <PES1UG24AM048>
Date:   1776748265

    Added file3

commit 49becd29aff19fad2b6b9ca34f7c139210375723bb0a6a3fe9a464d3764ccae8
Author: Apeksha Ashok <PES1UG24AM048>
Date:   1776748265

    Updated file1

commit f906b3599a6e578853b1c40fecdaffb4eaffaa9818fcfa39e7d4094246264198
Author: Apeksha Ashok <PES1UG24AM048>
Date:   1776748265

    Initial commita
```

```
aps@apekshaashok:~/AM048_PES_VCS$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/09/b576c2eb51b78fcf9b8493a0909522a75f8ee86df0e926312deeb5fbe7dfb1
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/38/ef0b23cc78fb48d733752af5e1b581dc076501ed5a5409adf781a19c01f655
.pes/objects/48/1407f7237f7bd749dca31e2d8422bafe211741d5556f007e0faf821ffc7179
.pes/objects/49/becd29aff19fad2b6b9ca34f7c139210375723bb0a6a3fe9a464d3764ccae8
.pes/objects/ca/bfedc966010a7ddc6d57502b7608c9c2768f51fc6d2e645c2d885d5bb2f8cf
.pes/objects/d1/191de3b7f7574404e0eb7e3869497e454112feb09527b93d1da0cb53169c85
.pes/objects/f9/06b3599a6e578853b1c40fecdaffb4eaffaa9818fcfa39e7d4094246264198
.pes/objects/fe/76325aa5521b207ebe01e12fd8e9e3abf030cacd5398e3744a3a56a81ad1bd
.pes/refs/heads/main
aps@apekshaashok:~/AM048_PES_VCS$ cat .pes/refs/heads/main
cat .pes/HEAD
cabfedc966010a7ddc6d57502b7608c9c2768f51fc6d2e645c2d885d5bb2f8cf
ref: refs/heads/main
```

```
aps@apekshaashok:~/AM048_PES_VCS$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 323deb9cd340... Initial commit

Log after first commit:
commit 323deb9cd340bee2915349acb638f67574a02f6dd85e7a6f73b9bd13261f7344
Author: Apeksha Ashok <PES1UG24AM048>
Date:   1776748385

    Initial commit
```

```
--- Second Commit ---
Committed: 5d92c84c525a... Update file.txt

--- Third Commit ---
Committed: ffaea9249b7d... Add farewell

--- Full History ---
commit ffaea9249b7d1e9222cfd56ecbbf64b1edcf2a5518fe7d487ae3a4d50a1006dc
Author: Apeksha Ashok <PES1UG24AM048>
Date:   1776748385

    Add farewell

commit 5d92c84c525a266280ae7ba3e11dfcf84bfe49ec3f83c7a60d017833c00288ef
Author: Apeksha Ashok <PES1UG24AM048>
Date:   1776748385

    Update file.txtemo CA"!

commit 323deb9cd340bee2915349acb638f67574a02f6dd85e7a6f73b9bd13261f7344
Author: Apeksha Ashok <PES1UG24AM048>
Date:   1776748385

    Initial commitQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
ffaea9249b7d1e9222cfd56ecbbf64b1edcf2a5518fe7d487ae3a4d50a1006dc
```

```
--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
ffaea9249b7d1e9222cfd56ecbbf64b1edcf2a5518fe7d487ae3a4d50a1006dc

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/32/3deb9cd340bee2915349acb638f67574a02f6dd85e7a6f73b9bd13261f7344
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/5d/92c84c525a266280ae7ba3e11dfcf84bfe49ec3f83c7a60d017833c00288ef
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/ff/aea9249b7d1e9222cfd56ecbbf64b1edcf2a5518fe7d487ae3a4d50a1006dc

=== All integration tests completed ===
```

Q5.1

To implement `pes checkout <branch>`:

- Check if branch exists in `.pes/refs/heads/`
- Update `.pes/HEAD` to point to that branch
- Read commit hash from branch file
- Load commit → get tree
- Replace working directory files with files from that tree
- Update index to match new state

Why complex:

Because it has to safely replace all files and avoid overwriting user changes.

Q5.2

To detect dirty working directory:

- For each file in index:
  - Compute current file hash
  - Compare with index hash
- If different → file is modified

Then compare with target branch:

- If file is modified locally and different in target branch → conflict

Q5.3

Detached HEAD means HEAD points directly to a commit, not a branch.

- New commits are created but not linked to any branch
- These commits can be lost if user switches branch

Recovery:

- Create a new branch pointing to that commit using its hash

Q6.1

Algorithm:

- Start from all branch heads
- Traverse all commits using parent links
- Mark all commits, trees, and blobs as reachable
- Store them in a hash set
- Delete objects not in this set

Data structure:

- Hash set (fast lookup)

Estimation:

For 100k commits → need to visit all reachable commits + trees + blobs
→ total can be hundreds of thousands of objects

Q6.2

- During commit, new objects are created but not yet linked
- If GC runs at same time, it may delete these objects

This breaks the repository

How Git avoids this:

- Uses locks
- Delays deletion
- Runs GC safely when no writes are happening